Behavioral Reputation Score

How RoundFi turns pool behavior into a portable, on-chain credit signal

RoundFi

2026-06-23

CONTENTS

> **Derived document.** This is a derivation of `docs/spec/MASTER-SPEC.md` (§5 Reputation engine, §6 Reputation metrics, §9 Adversarial model). The Master Spec is the single source of truth; if a number here disagrees with it, the Master Spec wins. Every constant is pinned to the deployed Jun 2026 source (`programs/roundfi-reputation/` , `services/indexer/`).

1. The thesis

RoundFi's product is **not** the savings pool — it is the **reputation** the pool produces. A rotating savings group (ROSCA) generates a stream of verifiable, financially-consequential events: paid on time, paid late, completed the round, defaulted. The reputation engine turns that stream into a **score**, a **tier** (L1–L4), and four **behavioral metrics**, and feeds the tier back into the protocol so that proven behavior lowers your cost of capital in the next pool.

The whole design rests on one separation: **receiving capital is not merit; keeping your obligations after receiving it is.** Everything below enforces that distinction on-chain.

2. The four-tier ladder

A member's tier sets the **stake** they must lock at join, as a fraction of the credit (`carta`) they can draw. The ladder is the user-facing reward — climb it and you free up capital.

Tier	Name	Stake (bps of credit)	Score threshold	Min completed pools	Identity
L1	Iniciante	5000 (50%)	0	0	—
L2	—	2500 (25%)	500	2	gate-configurable
L3	Veteran	1000 (10%)	2000	3	gate-configurable
L4	Elite	300 (3%)	5000	8	**always required**

Source: `roundfi-reputation/src/constants.rs` (`STAKE_BPS_LEVEL_*` , `LEVEL_*_THRESHOLD` , `LEVEL_*_MIN_CYCLES`).

An L4 Elite member risks **3%** of the credit where an L1 risks **50%** — a ~94% reduction in locked capital. That discount is the economic prize, which is exactly why the upper tiers are the ones worth gaming. The anti-farming walls in §4 are what make the discount safe to offer.

3. Score generation

The score is a single integer accumulated from attestations (see §5). The v1 schedule:

Event	Score Δ
On-time payment	+10
Completed pool	+50
Late payment	−100
Default	−500

A late payment costs ten on-time payments; a default wipes most of a pool's worth of progress. The asymmetry is deliberate — the score is meant to be **slow to earn and fast to lose**, because that is how trust actually works.

For **unverified** wallets the positive increments are halved (sybil-dampening); negative deltas are never dampened. A default zeroes the score outright and re-derives the tier immediately (a defaulter cannot re-enter the next pool at the cheaper L2/L3 stake — SEV-007).

4. Promotion is doubly-gated

`promote_level` (permissionless) advances a wallet to the **highest tier whose score AND completed-pools thresholds are both met** (`resolve_level`), then applies the identity cap (`cap_level_for_identity`). Two **independent** anti-farming walls must both fall:

4.1 The score wall

Score alone is **farmable**: an operator can spin up parallel one-member pools and harvest `+10` / `+50` in parallel. So score is necessary but never sufficient.

4.2 The completed-pools wall (the wall-clock defense)

`cycles_completed` only rises on a **`POOL_COMPLETE`** attestation — which fires once per pool the member **paid through to the end** — and each one carries a **30-day per-subject cooldown** (`MIN_POOL_COMPLETE_COOLDOWN_SECS`). This is the unbypassable defense: it cannot be parallelized, because the same wallet's pools are serialized by a 30-day wall clock.

The floors:

- **L2 needs 2 completed pools** (raised from 1 in **ECO-V52**, so the 4× stake discount is never reachable on a single self-dealt pool).
- **L3 needs 3.**
- **L4 needs 8** — roughly a multi-year honest history.

A score farm can buy points; it cannot buy time.

4.3 The identity hard floor

L4 Elite always requires `identity_verified`, independent of the configurable identity gate (`IDENTITY_HARD_FLOOR_LEVEL` , `cap_level_for_identity`). L2/L3 are governed by `IdentityGateConfig.required_min_level` (devnet `0` = open, mainnet `3` = verified-only). An unverified wallet is capped at **L3** even with the gate turned off — Elite is gated on proof-of-personhood by construction.

5. Attestation schemas (Pass-3 taxonomy)

Every consequential event emits an immutable on-chain attestation keyed by **subject = wallet**. The Pass-3 taxonomy (Jun 2026) is built around the "received vs kept" separation:

id	schema	emitter	score Δ	`cycles_completed`	polarity
1	`PAYMENT`	`contribute` (on-time)	+10	—	positive
2	`LATE`	`contribute` (late)	−100	—	negative
3	`DEFAULT`	`settle_default`	−500	—	negative
4	`POOL_COMPLETE`	`contribute` (final installment)	+50	**+1**	positive
5	`LEVEL_UP`	`promote_level`	0	—	informational
6	`PAYOUT_CLAIMED`	`claim_payout`	**0**	—	**neutral**

The critical correction (Pass-3): `PAYOUT_CLAIMED` — the event of *receiving* your carta — is **score-neutral and does not advance `cycles_completed`.** The `+50` and the cycle bump live on `POOL_COMPLETE` , which only fires on a member's **last** contribution of the pool — i.e. only after they have proven the pay-*after*-receiving discipline that is RoundFi's entire thesis. Verified live on devnet 2026-06-12 (pool `Ga2RwgSk…`): schemas 1, 6, and 4 emitted distinctly, `payout_claimed` landing in the neutral bucket.

6. The four behavioral metrics

The off-chain scorer (`services/indexer/src/reputationMetrics.ts`) projects the attestation history into four 0–100 metrics. Two are live; two ship as `null` until the canary calibrates them.

6.1 Reliability — *live*

A weighted average over the most recent `RELIABILITY_WINDOW` reliability-bearing events. `payout_claimed`, `pool_complete`, and `unspecified` carry **no** reliability weight. A fresh wallet defaults to `reliability = 0` (no evidence — not a penalty).

6.2 Punctuality — *live*

Derived from the average timing (paid-time minus due-time) over the recent payment window, mapped piecewise:

Avg timing	Punctuality
≤ 3 days early	100
early → on time	80 → 100
on time	80
up to 1 day late	80 → 60
1–7 days late	60 → 30
7–30 days late	30 → 0
> 30 days late	0

A **friction grace** treats any payment under 1h late as on-time (clock jitter, gas timing). Fresh-wallet neutral default is 80.

6.3 Commitment & Recovery — *pending*

- **Commitment** — depth of obligation kept (completed pools, post-payout discipline). Fed by `pool_complete`.
- **Recovery** — did a defaulter come back and rebuild? The single most important metric for the "second chance" thesis.

Both surface as `null` in the score API until the canary produces enough real default→recovery sequences to calibrate the weights. Shipping `null` rather than a fabricated number is deliberate: we do not publish a metric we cannot yet defend.

7. Why this resists farming

Attack	Why it fails
Score farming	The completed-pools floor + 30-day cooldown can't be parallelized; score alone never promotes.
Sybil	Each wallet must independently accrue completed pools over wall-clock time; reputation is per-wallet, non-transferable.
Elite farming	8 completed pools (~years of honest history) **plus** the identity hard floor make Elite the most-defended tier.
Goodhart	Promotion is gated on *consequential* on-chain events (real USDC at stake), not a soft signal — gaming it requires actually keeping obligations.

The portable output — a wallet's `{ tier, reliability, punctuality }` — is what an external lender (Kamino, MarginFi, a fintech) can consume to price credit for a borrower who would otherwise be invisible. That is the endgame the ladder exists to serve.

Cross-references: protocol mechanics → `02-technical-whitepaper`; program topology → `03-architecture-spec`; solvency math → `05-stress-lab-economic-model`. Full adversarial economics live in MASTER-SPEC §9.